

# **VOLUNCORE – PLATFORM FOR CONNECTING THE VOLUNTEERS**

A PROJECT REPORT

*Submitted by*

**JOSEPH AKASH C (962821205026)**

*a partial fulfillment for the award of the degree*

*of*

**BACHELOR OF TECHNOLOGY**

IN

INFORMATION TECHNOLOGY



UNIVERSITY COLLEGE OF ENGINEERING, NAGERCOIL

**ANNA UNIVERSITY: CHENNAI 600 025**

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## **BONAFIDE CERTIFICATE**

Certified that this report “**VOLUNCORE-PLATFORM FOR CONNECTING THE VOLUNTEERS**” is the bonafide work of “**JOSEPH AKASH C (962821205026)**” who carried out the project under my supervision.

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## ACKNOWLEDGEMENT

All praise, glory, and honor be to the Almighty Lord, the eternal source of all wisdom, for His divine presence and guidance that enabled us to work diligently and complete this project on time.

I take this opportunity to express my sincere gratitude to our respected Dean, **Dr. V. A. NAGARAJAN, Ph.D.**, for his constant support, valuable suggestions, and encouragement throughout the course of this project.

I am truly thankful to the Head of the Department and Project Coordinator, **Dr. A. RADHAKRISHNAN, M.E., Ph.D.**, Department of Information Technology, for his insightful guidance, timely support, and for providing us with the necessary time and resources to carry out this work.

It is also a great pleasure to extend my heartfelt thanks to my project guide, **Dr. J. VIJLA, M.Tech., Ph.D.**, Assistant Professor, IT Department, for her consistent mentorship, encouragement, and valuable inputs during each stage of the project.

I am also grateful to all the teaching and non-teaching staff members of the IT Department for their support and assistance. Finally, I would like to convey my deep appreciation to my parents and friends for their continuous encouragement, moral support, and motivation throughout this journey.

## **ABSTRACT**

The project is entitled as “VolunCore- Platform for Connecting Volunteers”. It is a web-based platform developed to connect the gap between Non-Governmental Organizations and individuals passionate about volunteering. These organizations often encounter difficulties in engaging and managing volunteers, while socially conscious individuals face challenges in finding and accessing meaningful opportunities. VolunCore addresses these challenges by offering a centralized, intuitive application that fosters seamless collaboration between organizations and volunteers. The platform enables organizations to register, create detailed activity listings, manage applications. Concurrently, volunteers can browse available activities, maintain their activity history, and receive timely updates. The system also includes secure authentication, profile management, announcement and feedback features and built with a responsive React frontend and a robust Spring Boot backend.

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## **LIST OF ABBREVIATIONS**

|       |                                    |
|-------|------------------------------------|
| NGO   | Non-governmental organization      |
| iOS   | iPhone Operating System            |
| UI    | User Interface                     |
| API   | Application Programming Interface  |
| JPA   | Java Persistence API               |
| UUID  | Universally Unique Identifier      |
| REST  | Representational State Transfer    |
| MySQL | My Structured Query Language       |
| IDE   | Integrated Development Environment |
| VANET | Vehicular Ad Hoc Network           |
| IoT   | Internet of Things                 |
| AI    | Artificial Intelligence            |
| QR    | Quick Response                     |
| GPS   | Global Positioning System          |

# CHAPTER 1

## INTRODUCTION

### 1.1 PROJECT OVERVIEW

In today's digital era, Non-Governmental Organizations (NGOs) play a vital role in addressing complex social issues, yet they often face significant challenges in recruiting, managing, and retaining volunteers. On the other hand, individuals who are eager to engage in meaningful social service frequently encounter fragmented or outdated platforms that hinder their ability to discover relevant volunteering opportunities. VolunCore is designed to bridge this gap—a centralized, user-friendly platform that fosters seamless collaboration between NGOs and volunteers. It empowers NGOs to post activities, manage volunteers, monitor participation, and collect feedback, while enabling volunteers to explore, apply for, and engage in causes aligned with their interests.

While digital transformation has permeated many sectors, volunteer coordination systems still struggle with inefficiencies such as overlapping tasks, poor communication, and difficulty in tracking contributions. Research shows that volunteer task coordination is often hindered by imbalanced workload allocation, inadequate transparency, and manual intervention, all of which reduce volunteer motivation and system performance [1]. To address this, VolunCore introduces intelligent activity management that allows NGOs to delegate responsibilities based on real-time availability, preferences, and historical participation, ensuring equitable distribution of work and reducing organizational overhead.

Volunteering, by its very nature, extends beyond operational contribution; it reinforces civic identity, fosters social cohesion, and nurtures a sense of shared purpose. Alsehaimi et al. emphasized the significance of structured volunteer systems that integrate both personal fulfillment and collective social impact [2].

Recognizing this, VolunCore not only matches volunteers to activities but also supports them in building their volunteering portfolio, tracking impact, and aligning with causes they care about. The platform is designed to promote sustained engagement by offering volunteers a sense of belonging and recognition within the ecosystem.

One of the core features that enhances volunteer participation is the inclusion of incentive mechanisms, even if non-monetary. Traditional systems often neglect the psychological and social motivations that drive volunteers, leading to disengagement over time. Decentralized or blockchain-inspired systems have demonstrated that transparent, traceable recognition—such as badges or digital certificates—can promote accountability and foster long-term involvement [3]. VolunCore implements this principle through a real-time acknowledgment and feedback system, where volunteers can earn verifiable credits and public appreciation for their contributions, which in turn strengthens trust and motivation.

A sustainable and effective volunteer management system must also ensure diversity and workload balance. As Samanta and Ghosh point out, a successful model integrates a diverse set of volunteers and prevents concentration of responsibilities among a few users, which can lead to fatigue or burnout [4]. VolunCore’s intelligent recommendation engine ensures activity matching not only by interest but also by skillset, gender, location, and availability. This multi-dimensional approach guarantees an inclusive experience for all participants and helps NGOs build more resilient and representative volunteer teams.

The technical core of VolunCore draws from contemporary models of serverless-assisted task coordination, which promote adaptability, scalability, and real-time scheduling [5]. By adopting this model, VolunCore dynamically adjusts activity assignments based on user status, availability, and feedback, without relying on centralized, static workflows. This allows the system to stay agile even

during periods of high demand or sudden fluctuations in user engagement. Furthermore, it reduces infrastructure overhead by leveraging event-driven execution, ensuring efficient resource utilization. This architecture also facilitates easier integration with third-party services, enhancing the platform's extensibility and responsiveness.

An essential aspect of platform performance is volunteer attendance and participation tracking, which helps NGOs assess effectiveness and make informed decisions. Inspired by systems that utilize adaptive job execution metrics in distributed networks, VolunCore introduces real-time attendance tracking and instant feedback capture [6]. Volunteers can check in to events, submit reflections, and rate their experiences, while NGOs can monitor trends and act promptly on feedback making the platform not just reactive but truly responsive.

The commitment dynamics between digital and offline volunteering also differ substantially. Ihm and Shumate observed that volunteers participating online require more deliberate reinforcement, structured updates, and active communication compared to their offline counterparts [7]. VolunCore addresses this through robust messaging features that include one-on-one chat between volunteers and organizations, broadcast notifications, and real-time updates about event changes or opportunities.

Although VolunCore is not a computing system in the traditional sense, its architecture borrows from the principles of infrastructure-assisted coordination, typically found in vehicular cloud and IoT platforms. In such systems, tasks are assigned dynamically based on availability and capability. VolunCore similarly considers volunteer load, skillsets, preferences, and location before recommending a role or assigning responsibility[8].

## 1.2 SCOPE OF THE PROJECT

The scope of VolunCore encompasses the end-to-end lifecycle of volunteering engagement—empowering organizations to post and manage social activities; enabling volunteers to discover, apply for, and participate in those initiatives; and facilitating seamless communication and feedback between both parties. The platform supports secure user registration and profile management for two primary roles (organization and volunteer), robust authentication, and role-specific dashboards, ensuring that each user sees only the actions and information relevant to their role. Organizations can create and publish activity listings with detailed descriptions, required skills, event schedules, location, and participation caps; monitor volunteer applications; and track attendance, task completion, and feedback. Volunteers can browse available activities based on their interests, location, or skills; apply to join causes; check upcoming commitments; and log their involvement and reflections after each activity. To enhance coordination, VolunCore keeps users informed about new opportunities, application status updates, task assignments, attendance confirmations, and feedback submissions. It also includes searchable listings of upcoming activities, volunteer history logs, and organizational reports for impact measurement and future planning.

Altogether, VolunCore eliminates manual follow-ups and communication gaps, promotes accountability, and fosters a vibrant, transparent ecosystem where volunteers and organizations can collaborate meaningfully. The system is designed to scale from local initiatives to large-scale community programs, ensuring that every effort counts towards creating measurable social impact with minimal friction.

## CHAPTER 2

### LITERATURE SURVEY

Based on the project overview and web application objectives, several key platforms, scholarly works, and research studies provide foundational insights and guide the development of this volunteer computing platform.

**Saleh and Shastry [1]** proposed a novel global task scheduling approach that significantly improves task distribution efficiency in volunteer computing systems. Their method strategically enhances resource utilization by dynamically allocating computational tasks to volunteers worldwide, addressing challenges related to scalability and heterogeneity inherent in distributed volunteer environments. This work emphasizes the critical importance of intelligent scheduling algorithms to maximize throughput and system reliability in volunteer-based platforms.

**Alsehaimi [2]** investigated the impact of volunteer work on social life quality through the lens of social work specialists. The study highlights not only the altruistic value but also the socio-psychological benefits volunteers gain, such as increased social connectedness and well-being. These findings underscore the necessity of fostering volunteer engagement and satisfaction, which are pivotal for long-term participation and retention in any volunteer computing or crowdsourcing platform.

**Al Ridhawi et al. [3]** introduced an innovative incentive mechanism that integrates blockchain technology into volunteer computing frameworks. By leveraging blockchain's decentralized ledger and smart contract capabilities, their system ensures transparency, traceability, and security of volunteer contributions

and rewards. This approach addresses trust issues commonly faced in volunteer management, making decentralized volunteer platforms more trustworthy and attractive for contributors.

**Samanta and Ghosh [4]** examined strategies for sustainable volunteer engagement within crowdsourcing paradigms, with a focus on retention and maintaining a diverse skill set among volunteers. Their research stresses the need for balancing volunteers' skills, interests, and motivation to create a balanced and effective workforce. They propose engagement models that support volunteers' continuous involvement, which is essential for sustaining the productivity of large-scale volunteer projects.

Building on this, **Samanta, Sethi, and Ghosh [5]** developed a serverless-assisted task assignment system that is both skill- and willingness-aware. Their approach intelligently matches tasks with volunteers based on individual capabilities and readiness, thereby fostering amicable collaboration and increasing overall efficiency. This system also highlights the advantages of serverless architectures for scalable and cost-effective volunteer management solutions.

**Nabil and Seghir [6]** proposed a social volunteer computing environment characterized by task-adapted scheduling policies. Their design adapts task allocation dynamically according to volunteer capabilities and system load, which optimizes resource use and responsiveness. This adaptability is crucial for managing diverse volunteer pools and fluctuating task demands, enhancing the robustness of volunteer computing platforms. By aligning tasks with the strengths and availability of volunteers, the system ensures higher efficiency and improved task completion rates.



**Ihm and Shumate [7]** explored differences in volunteer commitment across online and offline settings. Their findings reveal that volunteer engagement is context-dependent, requiring tailored communication strategies and motivational approaches for virtual environments. This research informs the design of engagement features that cater to both digital and physical volunteer participation, ensuring a more comprehensive and effective volunteer management system.

Finally, **Waheed et al. [8]** investigated an infrastructure-assisted job scheduling and coordination framework for volunteer computing integrated into Vehicular Ad Hoc Networks (VANET). Their study demonstrates how real-time infrastructure support can facilitate efficient task scheduling and coordination in highly dynamic and mobile environments. This highlights potential future extensions for volunteer computing platforms that operate in specialized or resource-constrained contexts.

Together, these works contribute significant theoretical and practical knowledge relevant to building a scalable, transparent, and engaging volunteer computing platform. They collectively emphasize the importance of optimized task scheduling, volunteer motivation and retention, security and trust mechanisms, as well as adaptive management strategies—all of which are integral to the success of the proposed system.

## **CHAPTER 3**

### **SYSTEM ANALYSIS**

#### **3.1 EXISTING SYSTEM**

Currently, NGOs and volunteers operate across disconnected systems that involve manual tracking, scattered communication, and limited transparency. Volunteer matching, feedback submission, and activity coordination are often done through disparate channels like WhatsApp groups, Google Forms, or isolated NGO websites. These fragmented workflows lead to inefficiencies in volunteer engagement, reduced retention, and lack of visibility into volunteer impact or organizational needs.

##### **3.1.1 METHODOLOGY**

###### **3.1.1.1 TRADITIONAL (OFFLINE) VOLUNTEERING PROCESS**

###### **Opportunity Discovery & Registration:**

Volunteers typically learn about opportunities through word-of-mouth, posters, or local awareness events. Registration, if done, is manual—volunteers fill out paper forms or submit interest via calls. NGO staff often maintain separate volunteer spreadsheets with no integration across teams.

###### **Task Assignment & Coordination:**

Activity assignments are communicated manually via text messages, phone calls, often resulting in missed updates or miscommunication. Volunteers frequently lack clarity on task expectations, specific schedules, required skills, or the duration of commitment. This ambiguity can lead to confusion, underperformance, or last-minute dropouts. The lack of centralized coordination also prevents efficient task delegation, especially during high-volume campaigns or events.

**Engagement & Communication:**

There's minimal two-way interaction. Volunteers cannot easily communicate progress or seek support, and NGOs rarely provide real-time updates. Follow-ups depend heavily on individual initiative.

**Feedback & Tracking:**

Volunteer contributions are rarely documented. No consistent feedback system exists—volunteers are often left unaware of the impact they made. NGOs do not systematically record feedback, leading to a lack of data-driven insights or improvement cycles.

**3.1.1.2 EXISTING ONLINE VOLUNTEERING PLATFORMS****Platform Fragmentation:**

Several platforms exist like VolunteerMatch, Idealist, or iVolunteer; however, each operates independently with distinct workflows. Volunteers must create separate accounts, search manually for tasks, and navigate different UI/UX patterns, resulting in a disjointed experience.

**Limited Task Mapping & Real-Time Coordination:**

Existing systems focus primarily on listing activities. There's limited logic to map tasks to volunteers based on skill, availability, or proximity. Real-time coordination is usually missing or third-party dependent.

**Lack of Customizability for Smaller NGOs:**

Most systems are built for large-scale NGOs and require a degree of technical setup. Small or rural organizations may not have access to trained personnel or the digital infrastructure to list opportunities, onboard volunteers, or manage feedback workflows effectively.

### **3.1.2 DISADVANTAGES**

#### **Disconnected Ecosystem:**

Volunteers, organizations, and tasks exist on separate planes—manual coordination and redundant registration lead to delays and loss of volunteer interest. This makes them unaware about the activity.

#### **No Intelligent Matching:**

Current systems do not assess volunteer skills, past involvement, or intent. This often results in misaligned assignments and lower volunteer satisfaction.

#### **Minimal Impact Transparency:**

Volunteers rarely receive structured feedback or progress updates. NGOs lack tools to showcase or analyze volunteer contributions, their performance, weakening credibility and retention.

#### **Barrier for Small-Scale Organizations:**

Without technical expertise or digital onboarding support, many local NGOs remain excluded from online volunteer platforms, losing access to critical human resources.

### **3.2 PROPOSED SYSTEM**

The proposed VolunCore platform is a fully integrated, role-based volunteer management system that connects NGOs and volunteers through a unified web portal. It features secure user registration and login, dynamic dashboards tailored to each role, structured volunteering activity listings, application tracking, and real-time announcements. It also sends the notifications to the volunteers by the email to alert them about the update or any changes in the place or activity.

Built with React.js on the frontend and Spring Boot on the backend, the platform streamlines key workflows: NGOs can post volunteering opportunities, track participant engagement, and submit feedback, while volunteers can browse opportunities, apply, communicate with organizers, and monitor their own progress.

### **3.2.1 METHODOLOGY**

The VolunCore system is developed using a layered, agile methodology that combines modern web technologies, secure REST APIs, and communication channels.

The user journey begins with a simple sign-up process where users choose their role as either NGO representatives or volunteers. NGOs are prompted to provide basic organizational details that personalize their dashboard and help establish credibility. Once registered, users can securely log in, with an email-based recovery option available in case of password loss.

NGOs gain access to an intuitive dashboard where they can create and manage volunteering activities. Posting an opportunity involves filling out a short form with details such as title, description, required skills, event date, and number of volunteers needed. Once submitted, the activity becomes publicly available to all volunteers in the system. As applications come in, the NGO dashboard reflects the real-time status of each applicant, including approval or rejection options and a simple interface to assign tasks or log performance.

Volunteers, upon logging in, are presented with a personalized dashboard listing all available volunteering opportunities. Each activity card includes details like required skills, date, location, and duration. Volunteers can apply to participate with a single click and receive instant confirmation or updates via real-time notifications. Once selected, they can view their assigned tasks and update status, ensuring accountability and progress tracking.

A key feature of the platform is the feedback module. After task completion, NGOs can submit structured feedback for each volunteer based on participation hours, attitude, punctuality, and task completion. This data is stored in a dedicated feedback table and can be viewed by volunteers to reflect on their performance. Over time, this forms a comprehensive participation record for each user.

Together, these components form a robust system where tasks are clearly defined, applications are handled efficiently, and all stakeholders—NGOs and volunteers—can interact transparently within a secure, centralized environment.

### **3.2.2 ADVANTAGES :**

#### **Role-Based User Experience:**

VolunCore delivers a clean, role-specific interface designed to match the unique workflows of NGO administrators and volunteers. NGO representatives can post, manage, and monitor volunteering activities, view participant applications, and submit structured feedback—all from a centralized dashboard. Volunteers, on the other hand, access curated lists of open opportunities, track their involvement, and manage communication, ensuring focused and intuitive interaction for each user group.

#### **Streamlined Activity Lifecycle Management:**

The platform offers end-to-end activity lifecycle control from opportunity creation to volunteer application, task completion, and feedback. NGOs have complete visibility into the status of each volunteer and can take timely action, while volunteers receive clear instructions and updates, reducing ambiguity and improving participation rates. The activity is created and visible to all the volunteers who are registered in the platform.

**Centralized Feedback and Progress Monitoring:**

VolunCore incorporates a structured feedback mechanism that enables NGOs to assess and log volunteer contributions after each activity. Volunteers can view their feedback history and track accumulated service hours, creating a sense of accomplishment and an evidence-based performance record that can be used for future reference or recognition.

**Real time Notifications:**

VolunCore includes a real-time notification system that keeps users promptly informed about important events such as new activity postings, application status updates, task assignments, and feedback submissions. This ensures that both volunteers and NGO administrators remain engaged and responsive without needing to constantly check the platform manually.

**Scalable and Maintainable Architecture:**

Developed using modern technologies like React.js and Spring Boot, and supported by a MySQL database, the VolunCore platform ensures clean code separation, RESTful service communication, and efficient data handling. This makes it easy to maintain, scale, and extend the platform as more NGOs and volunteers join.

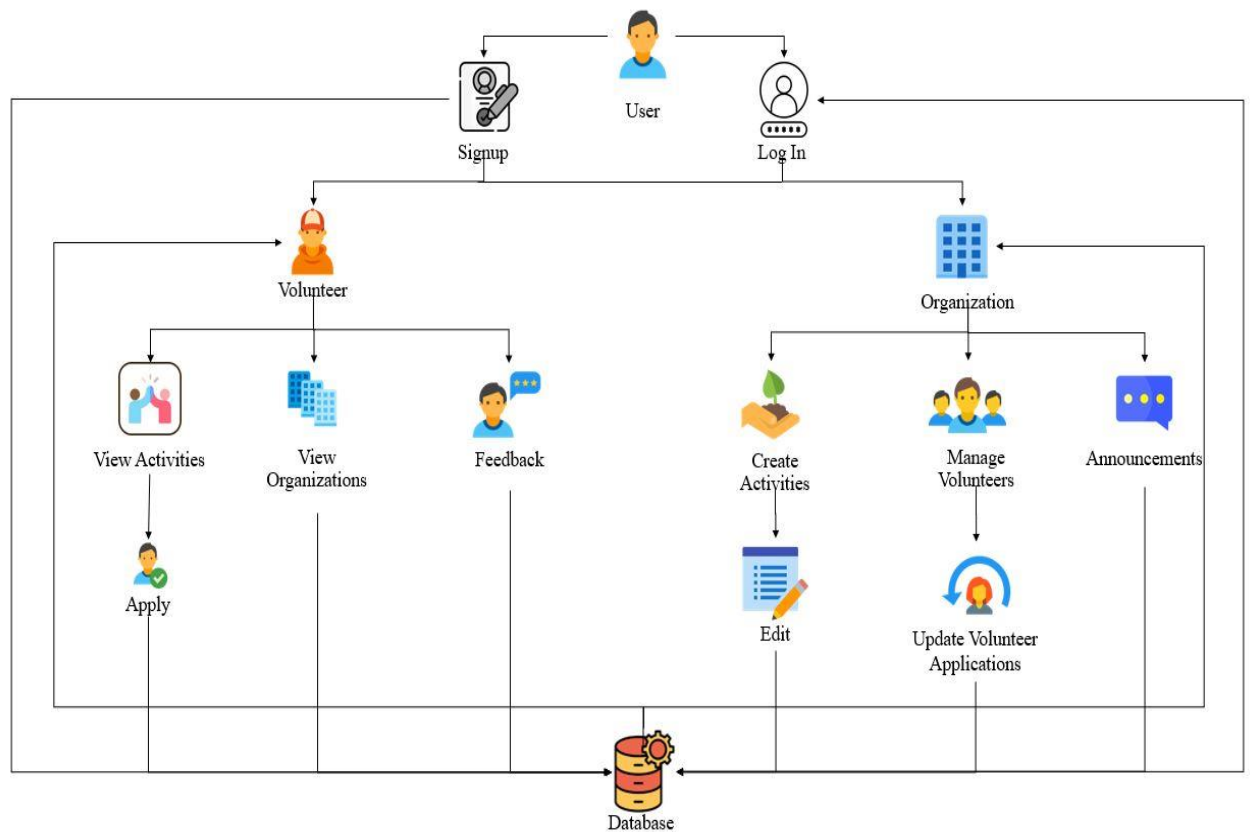
## CHAPTER 4

### SYSTEM DESIGN

The System Design gives the detailed diagrammatic representation of the proposed system with its methodology demonstrated in the previous chapter.

#### 4.1 SYSTEM ARCHITECTURE

Fig. No. 4.1 shows the system architecture of the VolunCore- Platform for connecting the volunteers.



**Fig. No. 4.1 System Architecture**



### 4.1.1 USER AUTHENTICATION

In the VolunCore platform, the authentication system facilitates user registration, login, and password reset functionalities specifically for two roles: Volunteer and Organization. The system stores data in separate tables for volunteers and organizations, ensuring role-specific data structures and workflows.

When a user wants to sign up, they are first presented with a tab-based interface on the frontend, allowing them to choose whether they are registering as a Volunteer or an Organization. Based on the selected tab, the corresponding registration form is displayed with the appropriate fields. They send a POST request to the `/api/volunteers` or `/api/organizations` endpoint. The data in the request varies depending on the user's role—whether they are a Volunteer, or an Organization representative. If any field is invalid, the system responds with a 400 Bad Request, listing all validation issues through a centralized global exception handler.

Once validation succeeds, the service layer processes the input. VolunCore application currently stores passwords in plain-text form for simplicity during the development phase. The system creates a new user record using a User entity, filling in fields like name, email, contact number, and user role. For organization users, additional fields such as `organizationName` and `contactPerson` are also captured. This data is persisted in the MySQL database using Spring Data JPA. A unique user ID is generated upon successful insertion.

Finally, the application responds with a 201 Created status, including a success message and basic user information like user ID, role, and—if applicable—the organization name. If the email already exists in the database (violating the uniqueness constraint), a 409 Conflict error is returned with a descriptive error message, handled consistently via the global exception handler.

To log in, users send a POST request to `/api/auth`, passing their email and password. The authentication service attempts to retrieve the user using the provided email. If no user is found, the system immediately returns a login failure response.

If a matching user is found, the input password is compared directly with the stored plain-text password. If the passwords match exactly, the system responds with a simplified user object containing essential details like user ID, role, and—for organizations—fields like `organizationName` and `organizationType`, along with a 200 OK status. If the password is incorrect, a custom `InvalidCredentialsException` is thrown and caught by the global exception handler, resulting in a 401 Unauthorized response with a message like "Invalid email or password."

If a user forgets their password, they can request a reset by sending a POST request to `/api/reset/forgot-password` with their email. The backend looks up the user and, if found, generates a unique reset token using a secure UUID. This token is stored in a separate password reset table along with an expiry time set to 48 hours from creation.

The backend then sends a password reset email using Spring's `JavaMailSender`, containing a link to the frontend application with the reset token embedded as a URL parameter. To complete the password reset, the user submits a POST request to `/api/reset/reset-password` along with the token and a new password. The system validates the token and checks that it hasn't expired. If the check passes, the user's plain-text password in the database is updated with the new one, and the reset token is deleted to ensure one-time use. A success message is returned, confirming that the password has been reset successfully. Also the old password is expired and not to be used again. The new password is updated in the database.

#### **4.1.2 VOLUNTEER ROLE**

After successful registration and login, volunteers access a personalized dashboard tailored to their engagement needs on the VolunCore platform. This dashboard provides an organized view of available volunteering opportunities, upcoming events, and the volunteer's active commitments. Volunteers can browse through a comprehensive list of activities posted by organizations and apply filters such as location, activity type, and dates to find the most suitable opportunities based on their interests and availability.

When a volunteer selects an opportunity to participate in, they submit an application through the platform. The backend validates these applications to avoid duplicates and ensures that the event or activity is still open for volunteers. Once accepted, these engagements are reflected in the volunteer's "My Activities" section, which helps them track their upcoming and past involvements, ensuring clear visibility into their volunteering schedules.

The platform also enables volunteers to provide feedback on activities they have completed. This feedback is essential for maintaining quality and accountability within VolunCore. Volunteers can rate events and organizations, and submit comments reflecting their experiences. These evaluations are stored in the database linked to the respective volunteer and event, contributing valuable insights for organizations to improve future opportunities and for other volunteers to make informed decisions.

Security and account management features allow volunteers to manage their profiles, update personal details, and reset passwords if needed. The password reset functionality involves sending a time-sensitive token to the volunteer's registered email, enabling secure password updates. While the current system stores passwords in plain text for development convenience, plans exist to implement secure password hashing in future versions.

Throughout the platform, role-based access control mechanisms ensure volunteers have permissions strictly aligned with their role and not to access the organization role by safeguarding sensitive information and maintaining operational integrity. The volunteer role within VolunCore is thus designed to offer an intuitive, secure, and streamlined experience, empowering individuals to effectively engage with community service activities and contribute positively.

#### **4.1.3 ORGANIZATION ROLE**

Organizations registered on the VolunCore platform play a crucial role in creating, managing, and promoting volunteering opportunities to attract and engage volunteers effectively. Upon logging in, organizations access a dedicated dashboard that offers an overview of their current and past activities, volunteer applications, and key engagement statistics, allowing them to manage their volunteer programs in one centralized interface.

Organizations can create new volunteering activities by submitting comprehensive details such as activity title, description, location, date, time, and the number of volunteers required. These activity postings are saved in the database and immediately made visible to volunteers browsing the platform. Organizations also retain full control to edit or cancel activities as necessary, ensuring that the information remains accurate and up to date.

A vital aspect of the organization's role is managing volunteer applications. When volunteers apply for an activity, organizations receive notifications and can review applicant details, accepting or rejecting applications through the platform. This streamlined application management fosters efficient communication and helps organizations select appropriate volunteers for each opportunity.

In addition to managing activities, organizations can post Announcements to communicate important updates, reminders, or general news related to their

volunteer programs. These announcements are displayed prominently on the organization's dashboard and can be viewed by all volunteers associated with their activities. This feature facilitates timely information sharing and strengthens engagement by keeping volunteers informed about changes, upcoming events, or urgent needs.

After activities conclude, organizations can provide feedback and ratings for volunteers who participated. This feedback system helps maintain a high standard of volunteer service and contributes to building reliable volunteer profiles within VolunCore, supporting future volunteer matching and recognition.

Account security and management for organizations include profile updates, password changes, and password reset functionality using secure email tokens to safeguard access. While the current development version stores passwords in plain text, plans for secure password hashing are intended for future releases to enhance protection.

In addition to these core capabilities, organizations on the VolunCore platform benefit from access to real-time analytics and reporting tools that offer valuable insights into their volunteer engagement trends. These tools allow organizations to track participation rates, measure the success of individual activities, and identify areas for improvement. Customizable reports can be generated to evaluate the effectiveness of recruitment strategies, volunteer retention, and overall program outcomes.

Overall, the organization role within VolunCore is designed to empower organizations with efficient tools to manage volunteering opportunities, oversee volunteer participation, and communicate effectively through announcements. This comprehensive functionality helps organizations maximize their community impact while maintaining strong, transparent connections with volunteers.

#### **4.1.4 ACTIVITY MANAGEMENT**

The Activity Management Module is one of the most important parts of the VolunCore platform. It allows organizations to create and manage volunteering tasks, and lets volunteers easily find and apply for opportunities that match their interests and skills.

For organizations, this module provides a simple form where they can create a new activity by entering details like the activity name, description, date and time, location (physical or online), required skills, and the number of volunteers needed. Once an activity is posted, it becomes visible to all volunteers on the platform.

Organizations can view a list of people who apply for their activities and decide who to accept or reject. They can also edit the activity information if needed or close it when enough volunteers have joined. This helps organizations keep everything organized and up to date.

For volunteers, the module offers an easy way to browse and filter activities based on their location, availability, interests, or skills. Each activity shows clear details so they can decide if they want to join. Volunteers can apply for an activity with just one click, and they can track the status of their application (like pending, accepted, or rejected). Once accepted, volunteers can view their upcoming activities in their personal dashboard. If something changes, they also have the option to withdraw from an activity before it starts.

All this information is stored securely in the platform's MySQL database. Each activity is linked to the organization that posted it, and every volunteer application is recorded with proper references to ensure nothing is lost or mismatched. The system also checks that all important fields are filled in correctly to avoid any confusion or errors.

Organization admins can see overall statistics—such as which activities are most popular, how many volunteers are participating, and which organizations are most active. This helps improve the platform and ensure that it runs smoothly.

To keep users informed, the system sends notifications when there are updates—like when a volunteer is accepted, or when an activity is about to start. These reminders can appear inside the platform or be sent through email if needed.

Activity Management module makes it easy for organizations and volunteers to connect, communicate, and collaborate. It supports smooth coordination, encourages participation, and plays a big role in making VolunCore an effective platform for community service.

#### **4.1.5 FEEDBACK SYSTEM**

The feedback system in VolunCore is a key component designed to foster continuous improvement and trust between volunteers and organizations. After the completion of a volunteering activity, both volunteers and organizations have the opportunity to provide constructive feedback on their experience. This two-way feedback mechanism ensures accountability, enhances the quality of volunteering opportunities, and helps build a reliable community reputation on the platform.

For volunteers, the feedback system allows them to share their thoughts on the organization, the clarity of activity instructions, support received during the task, and overall satisfaction with the volunteering experience. Volunteers can submit ratings and comments through an easy-to-use interface on their dashboard, which is then stored securely in the database. This feedback helps

organizations identify areas for improvement, recognize successful efforts, and maintain high standards of volunteer engagement.

On the organization side, administrators can evaluate volunteers based on their punctuality, commitment, skill application, and attitude during the activity. Organizations provide ratings and detailed comments that contribute to the volunteer's profile, assisting future organizations in making informed decisions when selecting volunteers for upcoming activities. This promotes a culture of reliability and motivates volunteers to maintain high levels of performance.

All feedback entries are recorded in a dedicated feedback table within the MySQL database, linked to both volunteer and organization entities through foreign keys to ensure data integrity. The system enforces validation rules to guarantee that feedback is complete and meaningful, such as requiring a minimum rating and non-empty comments where appropriate. Administrators can view aggregated feedback reports and trends on their dashboard, enabling data-driven decisions and strategic improvements.

To maintain a positive and respectful community environment, VolunCore includes mechanisms to flag inappropriate or abusive feedback. Such reports trigger reviews by platform moderators who can take necessary actions, including feedback removal or user warnings, ensuring the platform remains a safe space for all users.

To further encourage active engagement with the feedback system, VolunCore incorporates incentive mechanisms that recognize consistent and constructive contributors. Volunteers who regularly receive positive feedback are awarded digital badges and visibility boosts on their profiles, increasing their chances of being selected for future activities. Similarly, organizations with a strong history of fair and timely feedback are highlighted as top-rated hosts, enhancing their credibility within the community. These recognition features



serve as positive reinforcement, motivating users to participate in the feedback process with sincerity and integrity.

In summary, the feedback system in VolunCore enhances transparency, encourages quality contributions, and strengthens the relationship between volunteers and organizations. By enabling structured and thoughtful evaluations, it supports continuous growth and trust within the volunteer community, which is essential for the platform's long-term success.

#### **4.1.5 ANNOUNCEMENT SYSTEM**

The announcement system in VolunCore serves as a central communication channel where organizations can broadcast important updates, upcoming event information, or urgent notices to their registered volunteers. This feature helps maintain timely and effective communication, ensuring volunteers stay informed and engaged with the latest organizational activities and opportunities.

Organizations create announcements through a dedicated interface in their dashboard, where they can compose messages, set publication dates, and optionally attach relevant documents or links. These announcements are stored in a database table linked to the respective organization's profile, allowing for easy retrieval and management. The system supports scheduling, so announcements can be pre-written and published automatically at specified times, enabling organizations to plan communications strategically.

On the volunteer side, announcements are displayed prominently within their personalized dashboard and notification area. Volunteers receive clear visibility of new announcements related to the organizations they are affiliated with or have previously volunteered for. This targeted delivery ensures that

volunteers receive relevant information without unnecessary noise, increasing the likelihood of participation and responsiveness.

To keep volunteers engaged, the announcement system also sends reminders about important updates or upcoming events. This helps make sure volunteers don't miss any key information and can plan their participation accordingly. By keeping communication regular and timely, the platform strengthens the connection between organizations and their volunteers.

To enhance user experience, the announcements system incorporates features such as read/unread status tracking, so volunteers can quickly identify new messages. Additionally, volunteers can filter or search announcements by date, organization, or keyword, facilitating efficient access to information.

Administrators have access to analytics that provide insights into announcement reach and volunteer engagement, such as the number of views and responses. This data helps organizations refine their communication strategies and tailor announcements to maximize impact.

Overall, the announcements system in VolunCore plays a vital role in connecting organizations and volunteers, promoting transparency, timely updates, and stronger community involvement. It ensures that crucial information flows seamlessly, contributing to a well-coordinated and responsive volunteer network. By keeping volunteers informed and engaged, the system helps improve participation rates and supports the successful execution of volunteer activities. This continuous communication fosters trust and collaboration, which are essential for building a thriving volunteer community.

## CHAPTER 5

### SYSTEM SPECIFICATION

The project is web application based and it has some of the hardware and the software requirement to fulfill the overall scope of the project.

#### 5.1 SOFTWARE REQUIREMENTS

- **IntelliJ IDEA:** It is a widely-used integrated development environment (IDE) favored by developers for writing, testing, and managing code efficiently.
- **Visual Studio Code:** VS Code is a free, lightweight, and powerful code editor developed by Microsoft.
- **MySQL Workbench:** It is a graphical tool used for managing MySQL databases.
- **Postman API:** It is a popular API testing tool used by developers to send requests and receive responses from APIs

#### 5.2 HARDWARE REQUIREMENTS

- **CPU and Memory:** A server with at least 2 CPU cores and 4GB of RAM.
- **Processor:** Intel AMD Processor with Radeon Graphics - 2.30 GHZ.
- **Storage:** 10 GB free disk space.
- **Network:** Internet access for package dependencies , email delivery, and API calls.

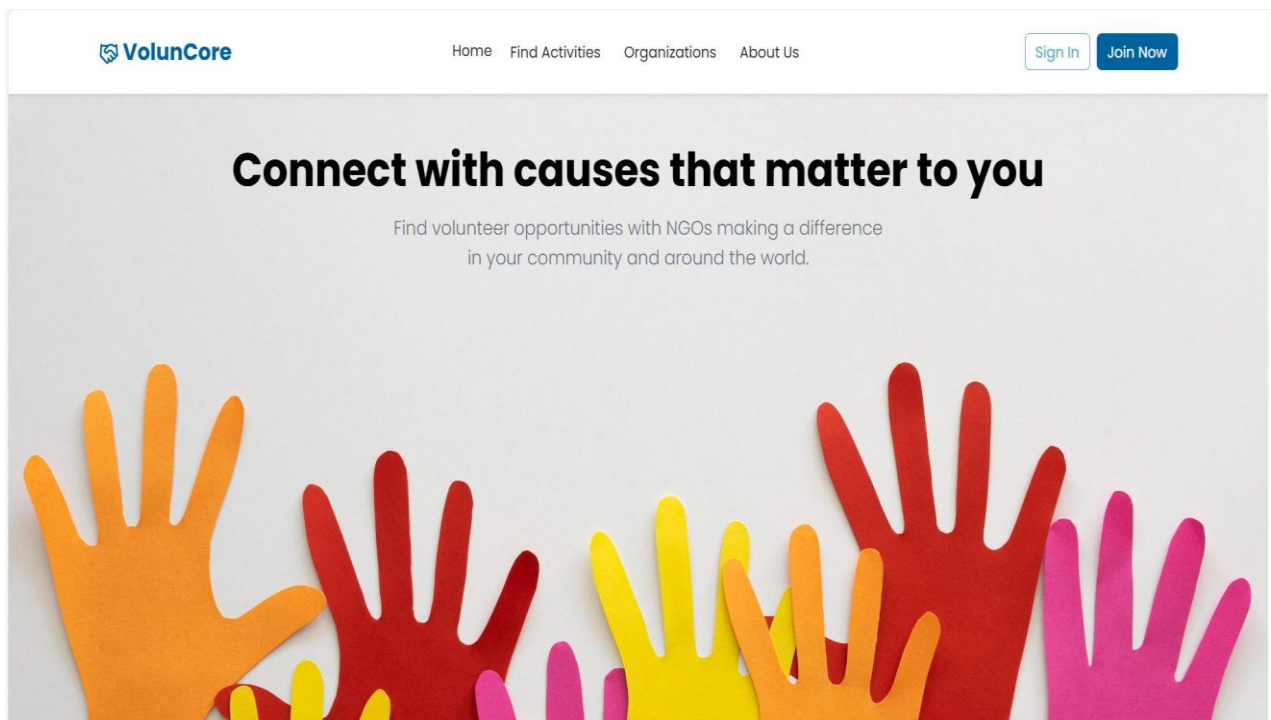
## CHAPTER 6

### IMPLEMENTATION AND RESULT

Based on proposed system methodology a web application is implemented successfully and demonstrated below.

#### 6.1 HOME PAGE

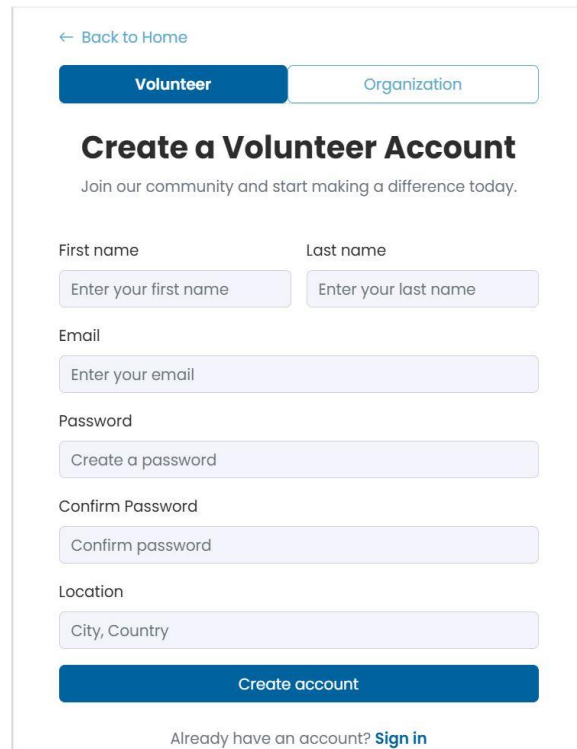
The home page is the front and the main page of the application which contains the general information about the system. This page will direct the user to the account creation page and the login page which is shown in Fig. No. 6.1. It also highlights the platform's purpose and key features, offering a clear starting point for new and returning users. Additionally, it provides intuitive navigation to explore other sections of the platform.



**Fig. No. 6.1 Home Page**

## 6.2 USER REGISTRATION PAGE

The user registration page guides new users through role-specific registration in a smooth flow is shown in Fig. No. 6.2.



The image shows a user registration form for the VolunCore platform, specifically for the 'Volunteer' role. At the top, there is a navigation bar with a 'Back to Home' link and two tabs: 'Volunteer' (selected) and 'Organization'. Below the tabs, the heading 'Create a Volunteer Account' is displayed, followed by the subtext 'Join our community and start making a difference today.' The form consists of several input fields: 'First name' and 'Last name' (each with a placeholder 'Enter your first name' and 'Enter your last name' respectively), 'Email' (placeholder 'Enter your email'), 'Password' (placeholder 'Create a password'), 'Confirm Password' (placeholder 'Confirm password'), and 'Location' (placeholder 'City, Country'). A large blue 'Create account' button is positioned below the location field. At the bottom, there is a link that says 'Already have an account? Sign in'.

**Fig. No. 6.2 User Registration page**

In the VolunCore platform, during registration, users begin by selecting their role as either a Volunteer or an Organization. Based on this selection, the form dynamically adjusts to collect the appropriate information. For Organizations, fields such as organization name, in-charge person's name, and contact number are displayed, while Volunteers are prompted to provide personal details like full name, email, and phone number. All users are required to enter and confirm a password. The Location is used to store the location in the profile section.

## 6.3 LOGIN PAGE

The Login Page component presents a clean, mobile-friendly form that lets users sign into VolunCore is shown in Fig. No. 6.3.

← Back to Home

### Sign in

Enter your email and password to access your account

I am signing in as:

**Volunteer** Organization

Email address

Enter your email

Password

Enter your password

[Forgot password?](#)

**Sign In**

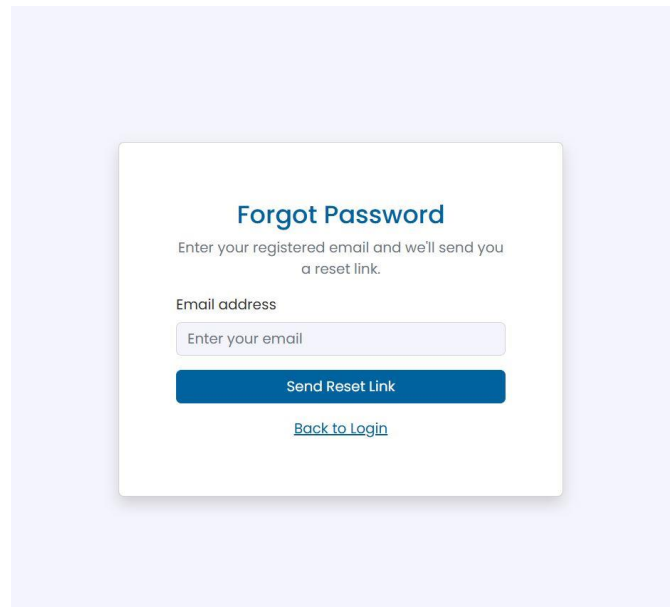
Don't have an account? [Sign up](#)

**Fig. No. 6.3 Login Page**

In the VolunCore platform, the login interface prompts users to enter their email and password, both of which are mandatory. Behind the scenes, the application verifies these credentials against the records stored in the system. If the entered details are correct, the user is successfully authenticated, and their profile information is saved locally in localStorage to maintain the login session and avoid repeated sign-ins. Upon successful login, the user is automatically redirected to the appropriate dashboard based on their role—Volunteer or Organization - ensuring a tailored experience.

## 6.4 FORGOT PASSWORD PAGE

The Forgot Password page is shown in the Fig. No. 6.4 which guides the user type their email and press the “Send Reset Link” button.

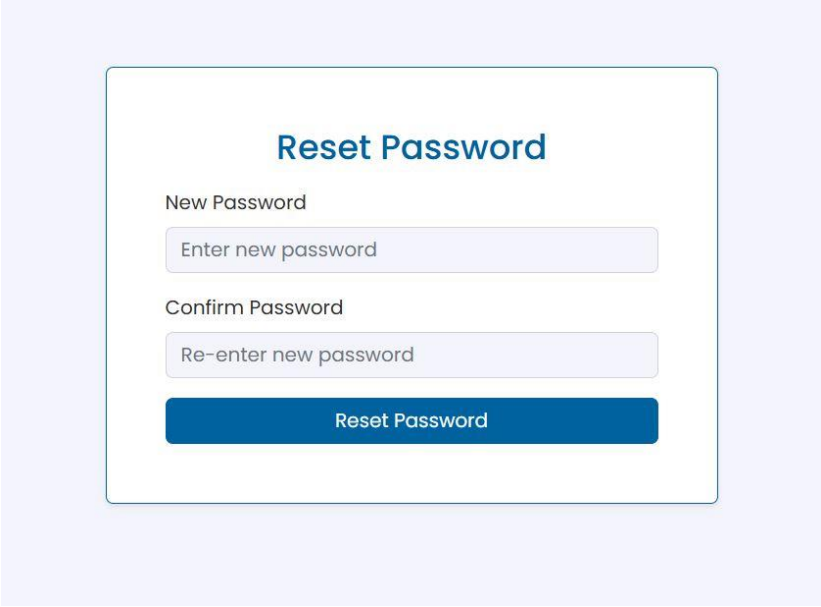
A screenshot of a web form titled "Forgot Password" in blue text. Below the title, a message reads: "Enter your registered email and we'll send you a reset link." The form includes a label "Email address" above a light blue input field containing the placeholder text "Enter your email". Below the input field is a solid blue button with the text "Send Reset Link" in white. At the bottom of the form is a blue hyperlink labeled "Back to Login". The entire form is centered on a light blue background.

**Fig. No. 6.4 Forgot Password Page**

The system then checks the address and, if it is found, sends an email containing a link. The user sees a message on screen confirming the reset link was sent. Next, the user opens their email inbox, finds the message, and clicks the reset link. This opens a secure form where the user can enter a new password. After submitting, the system updates the login credentials and confirms success. The user can then log in with the new password. This process helps protect user accounts by ensuring only the rightful owner can change the password, and it improves overall account security and user experience. It sends the email with the token which will expiry after the session is over.

## 6.5 RESET PASSWORD PAGE

The Reset Password page is shown in the Fig. No. 6.5, guides the user through choosing a new secret word when they click a dedicated link.

The image shows a 'Reset Password' form centered on a light purple background. The form is enclosed in a white box with a thin blue border. At the top of the box, the title 'Reset Password' is displayed in a bold, blue font. Below the title, there are two input sections. The first section is labeled 'New Password' in a small, grey font, followed by a light purple input field with the placeholder text 'Enter new password'. The second section is labeled 'Confirm Password' in a small, grey font, followed by a light purple input field with the placeholder text 'Re-enter new password'. At the bottom of the form, there is a solid blue button with the text 'Reset Password' in white, centered on it.

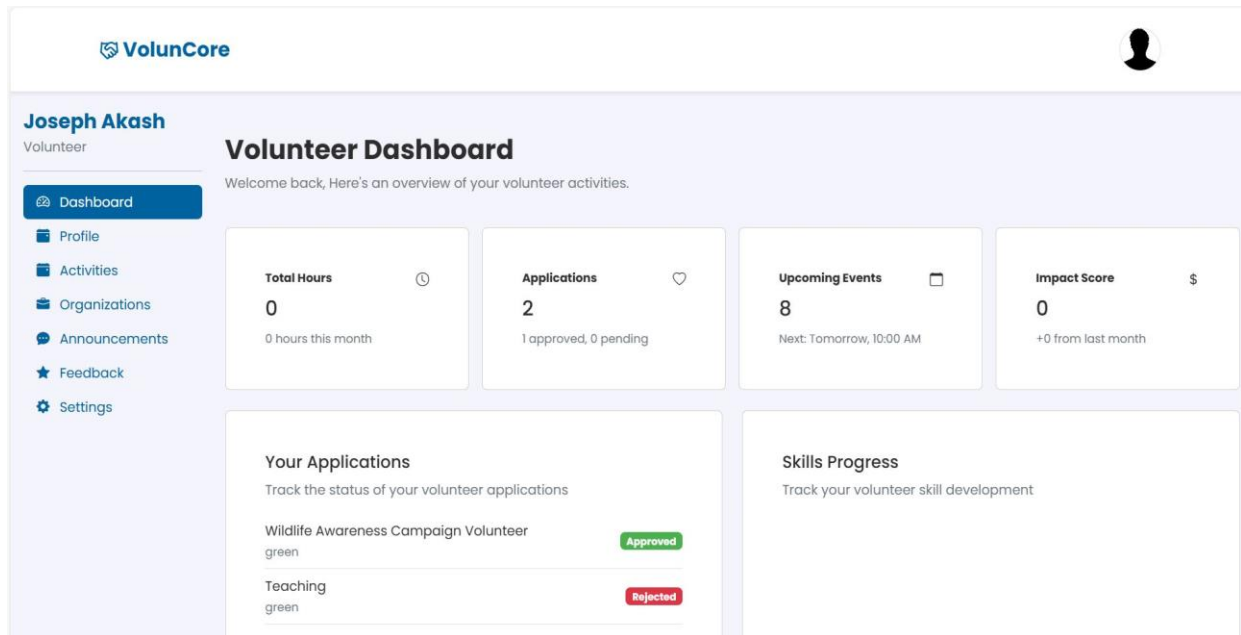
**Fig. No. 6.5 Reset Password Page**

When a user accesses the password reset page through a link, the system first verifies the reset token included in the URL. If the token is missing, invalid, or has expired, the user is immediately shown an appropriate error message. If the token is valid, the user is prompted to enter a new password and confirm it by typing it again. The system ensures that both passwords match and meet the required length and format criteria. Once validated, clicking the “Reset Password” button sends the token along with the new password to the server. If the operation is successful, the user receives a confirmation message indicating that the password has been successfully updated.



## 6.6 VOLUNTEER DASHBOARD

The volunteer dashboard is shown in Fig. No. 6.6 which show the analytics of the volunteers and also status of their applications.

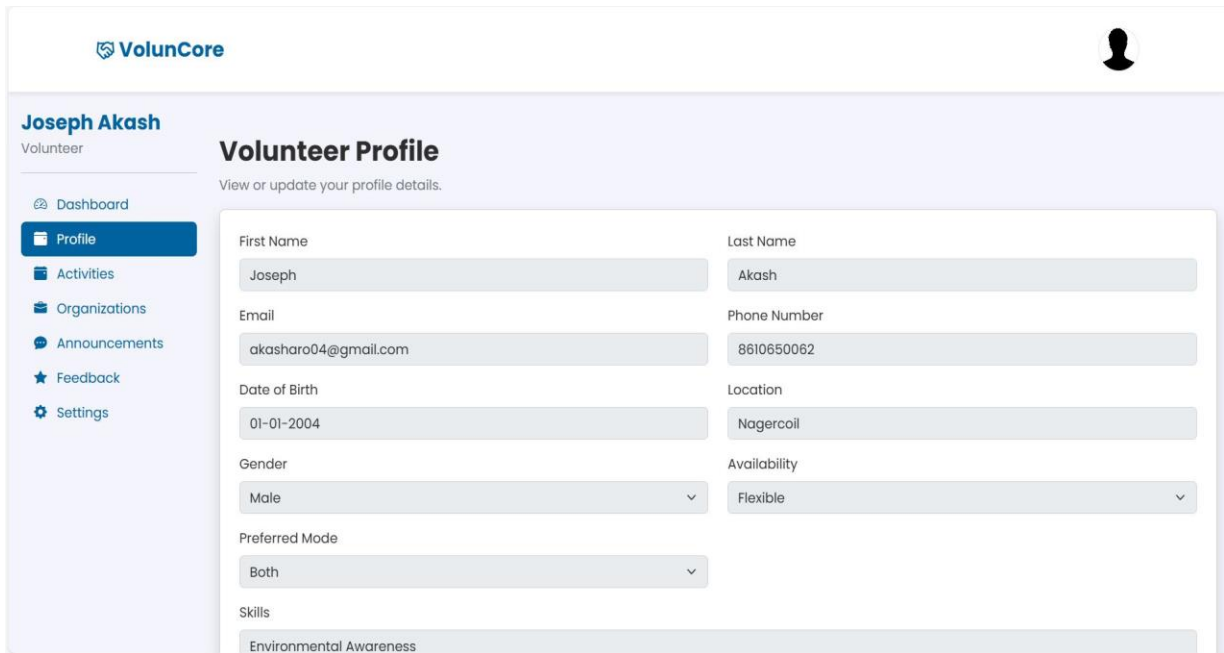


**Fig. No. 6.6 Volunteer Dashboard**

The dashboard provides a clear, user-friendly interface displaying upcoming tasks, active opportunities from partnered organizations. Volunteers can browse available volunteering activities, view detailed descriptions, and claim tasks that match their interests or skills. The system ensures that each opportunity claimed is tracked, allowing the volunteer to monitor their task status, completion history, and any pending responsibilities. Additionally, the dashboard showcases key statistics, such as total hours volunteered, completed tasks, and earned badges or recognitions.

## 6.7 VOLUNTEER PROFILE

The Volunteer Profile is shown in the Fig. No. 6.7, this page shows the details of the volunteer , if it is the new volunteer the page needs to update.



The screenshot displays the 'Volunteer Profile' page on the VolunCore platform. The header includes the VolunCore logo and a user profile icon. The left sidebar shows navigation options: Dashboard, Profile (selected), Activities, Organizations, Announcements, Feedback, and Settings. The main content area is titled 'Volunteer Profile' with a subtitle 'View or update your profile details.' Below this, there are several input fields for personal information: First Name (Joseph), Last Name (Akash), Email (akasharo04@gmail.com), Phone Number (8610650062), Date of Birth (01-01-2004), Location (Nagercoil), Gender (Male), Availability (Flexible), Preferred Mode (Both), and Skills (Environmental Awareness). Each field is represented by a light gray input box with a dropdown arrow on the right side.

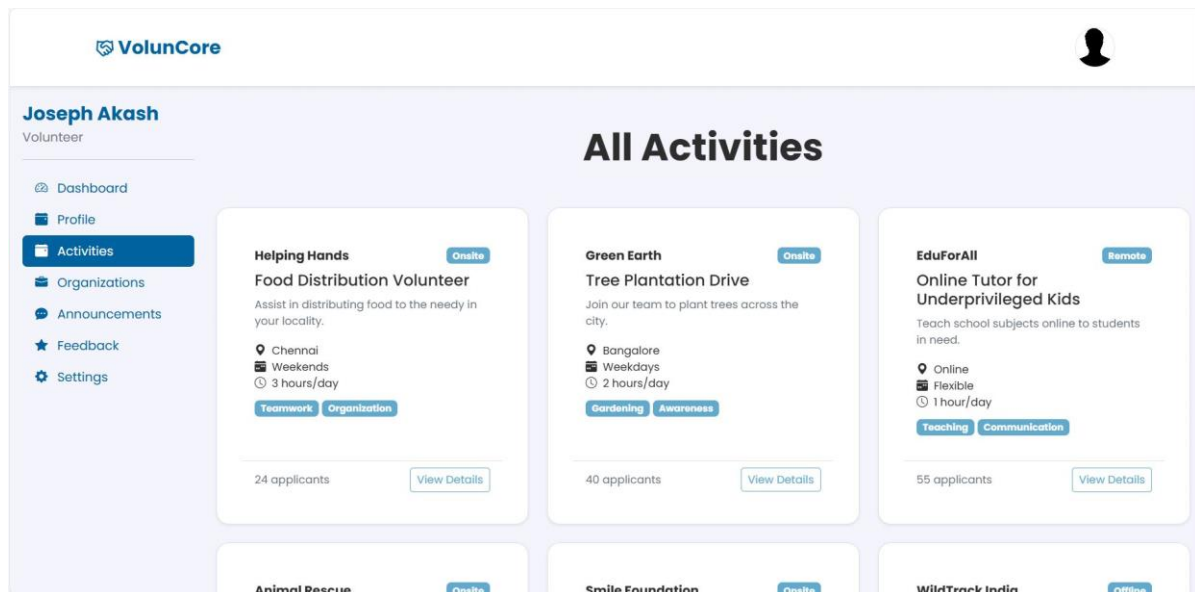
| Field          | Value                   |
|----------------|-------------------------|
| First Name     | Joseph                  |
| Last Name      | Akash                   |
| Email          | akasharo04@gmail.com    |
| Phone Number   | 8610650062              |
| Date of Birth  | 01-01-2004              |
| Location       | Nagercoil               |
| Gender         | Male                    |
| Availability   | Flexible                |
| Preferred Mode | Both                    |
| Skills         | Environmental Awareness |

**Fig. No. 6.7 Volunteer Profile**

The volunteer profile serves as a comprehensive summary of a volunteer's personal details. Upon successful registration and login, each volunteer is provided with a dedicated profile page where they can view and manage their personal information such as name, email address, contact number, and location. This section also includes editable fields, allowing volunteers to update their details when necessary, ensuring that their contact and availability information remains current for organizations to reference.

## 6.8 ACTIVITIES LIST

The Activities List shown in the Fig. No. 6.8, the Volunteer can see the available list of activities created by the organizations.

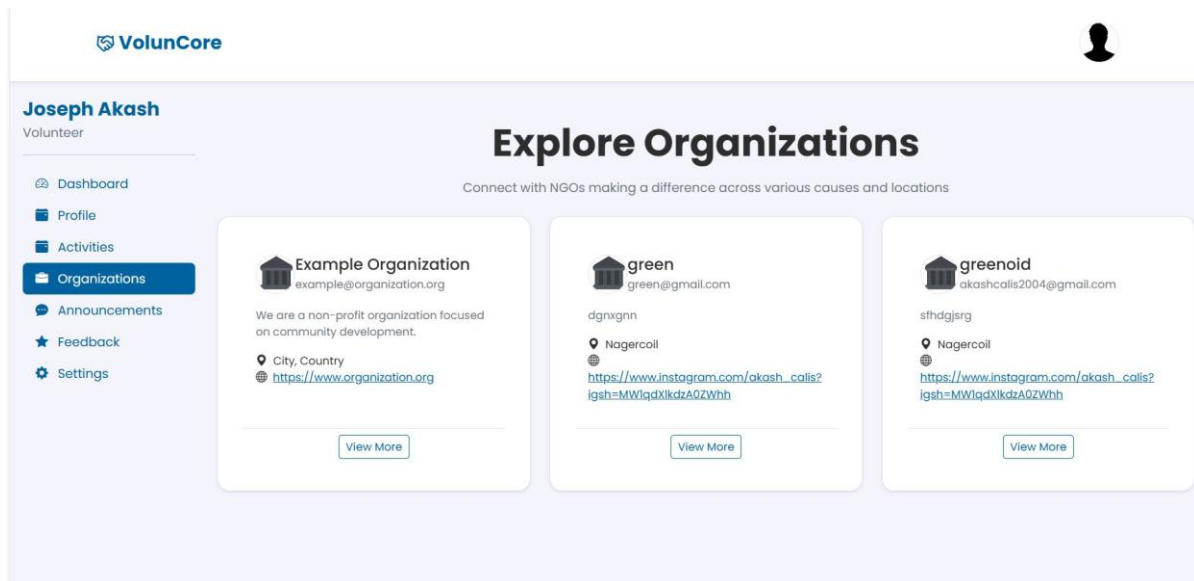


**Fig. No. 6.8 Activities List**

The Activities List acts as the central hub where volunteers can explore and discover opportunities posted by various registered organizations. This section presents a dynamic and scrollable list of all available community tasks, donation drives, awareness campaigns, and volunteering initiatives that are currently open for participation. Each activity card or row in the list typically includes essential information such as the activity title, a brief description, the organizing group's name, location, date and time of the event, and the number of volunteers required.

## 6.9 ORGANIZATIONS LIST

The Organization List shown in the Fig. No. 6.9, the Volunteer can see the available list of organizations.

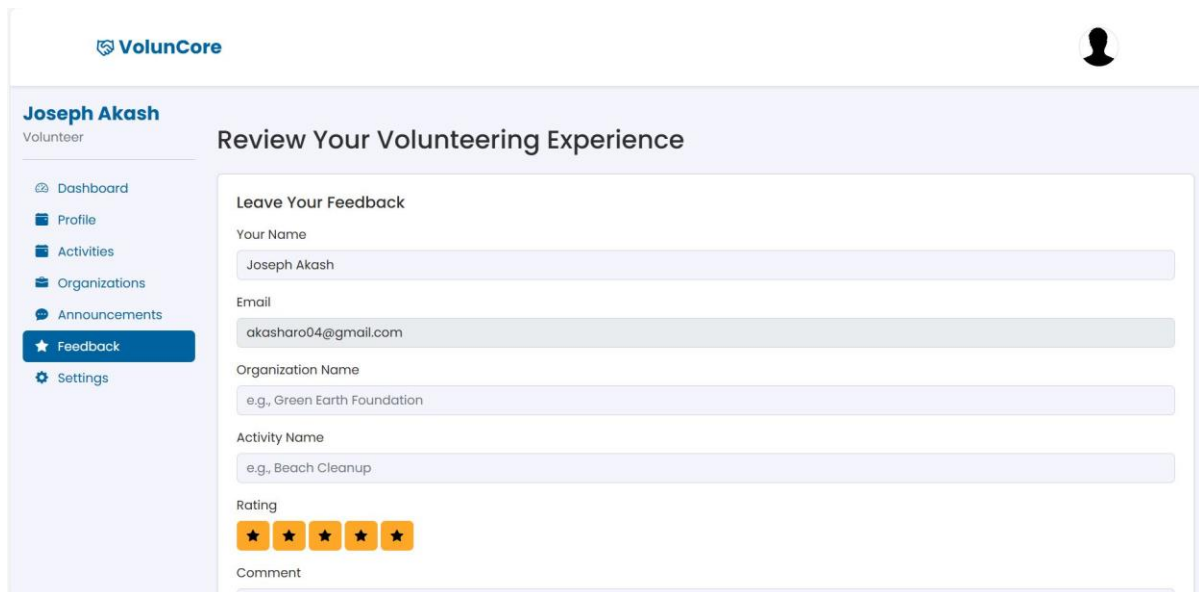


**Fig. No. 6.9 Organizations List**

The Organizations List serves as a comprehensive directory where volunteers can browse and connect with NGOs on the platform. This section presents a scrollable, card-based layout showcasing all organizations that are registered. Each organization card includes essential information such as the organization's name and location. To enhance user engagement, each card features a "View More" button that allows volunteers to access detailed information about the organization, including its full profile and contact details. This functionality enables volunteers to make informed decisions, explore organizations that align with their interests.

## 6.10 VOLUNTEER FEEDBACK PAGE

The Volunteer Feedback Page is shown in the Fig. No. 6.10, which shows the form for the feedback and the list of submitted feedbacks.



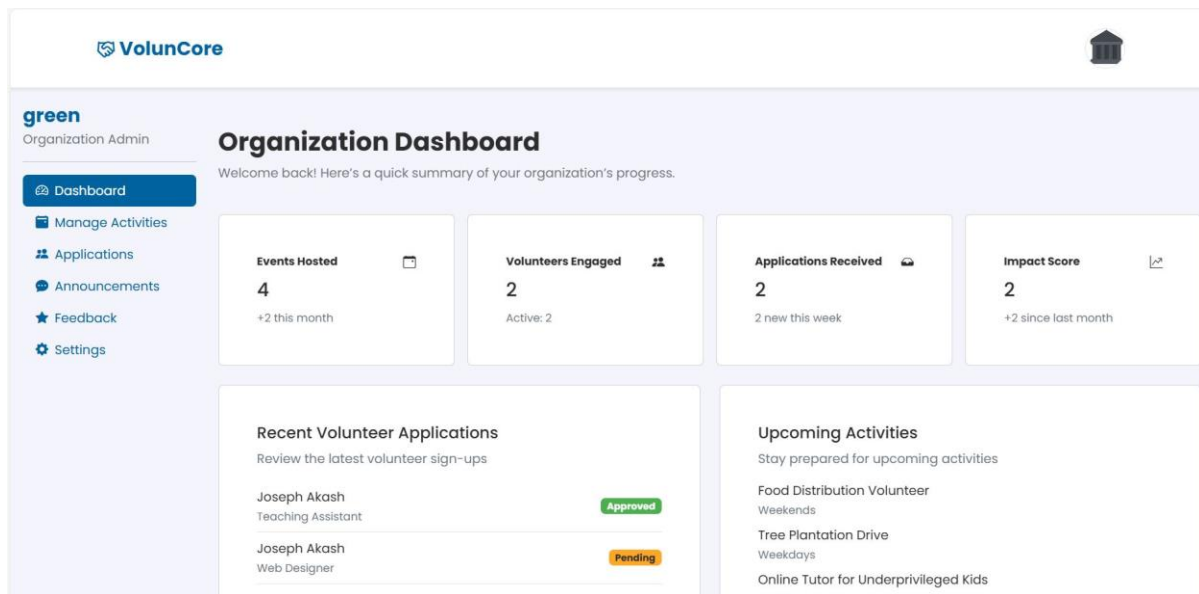
The screenshot displays the 'VolunCore' interface. At the top left is the 'VolunCore' logo, and at the top right is a user profile icon. Below the logo, the user's name 'Joseph Akash' and role 'Volunteer' are shown. A sidebar on the left contains navigation links: Dashboard, Profile, Activities, Organizations, Announcements, Feedback (highlighted with a star icon), and Settings. The main content area is titled 'Review Your Volunteering Experience'. It features a 'Leave Your Feedback' form with the following fields: 'Your Name' (filled with 'Joseph Akash'), 'Email' (filled with 'akasharo04@gmail.com'), 'Organization Name' (placeholder: 'e.g., Green Earth Foundation'), and 'Activity Name' (placeholder: 'e.g., Beach Cleanup'). Below these is a 'Rating' section with five orange star icons, the first three of which are filled. A 'Comment' text area is located at the bottom of the form.

**Fig. No. 6.10 Volunteer Feedback Page**

The Volunteer Feedback Page in the VolunCore platform provides a dedicated space where volunteers can view and manage feedback they've received from organizations after completing tasks or participating in events. This page plays a crucial role in building trust, encouraging improvement, and recognizing the efforts of volunteers. It offers a clear and organized view of ratings and comments, helping volunteers understand their strengths and areas for growth. By regularly accessing this feedback, volunteers can enhance their skills and increase their chances of being selected for future opportunities.

## 6.11 ORGANIZATION DASHBOARD

The Organization Dashboard is shown in the Fig. No. 6.11, which shows the analytics of the Organizations, received applications and the list of upcoming activities.

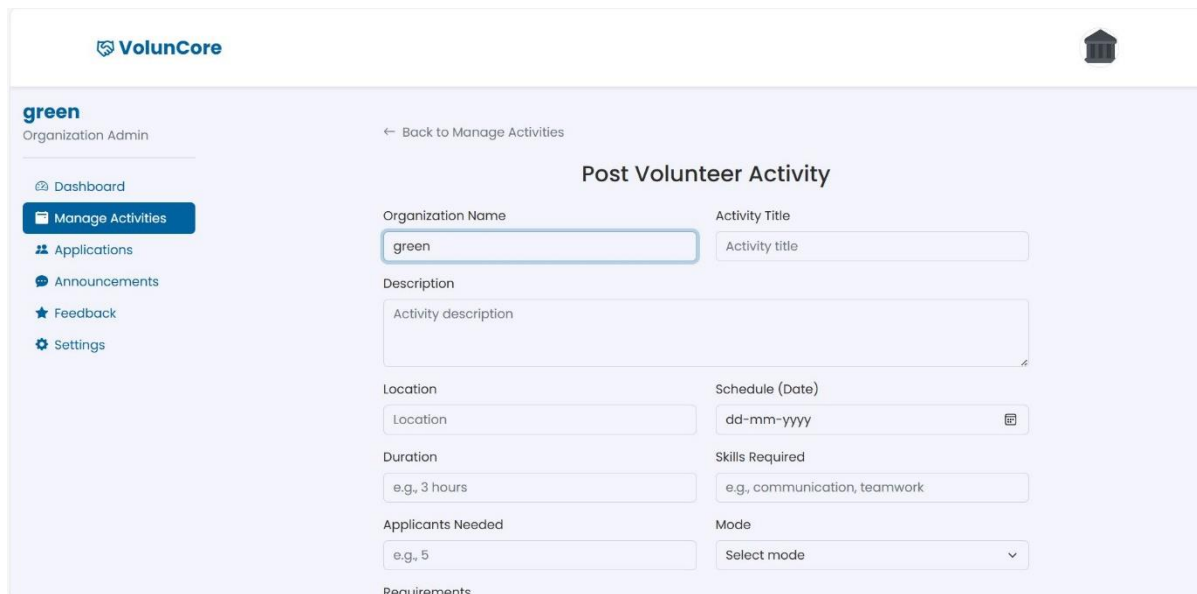


**Fig. No. 6.11 Organization Dashboard**

The Organization Dashboard serves as the central control panel for organizations to manage their activities, volunteers, and community engagement efficiently. It contains the analytics about the organization and its performance. Upon logging in, the organization is greeted with a personalized dashboard that provides a quick overview of key statistics such as the number of active volunteers, posted activities, upcoming activities. The dashboard also highlights recent volunteer applications and feedback received, allowing organizations to take timely actions. This centralized view helps organizations stay organized and make informed decisions to enhance their volunteer programs.

## 6.12 MANAGE ACTIVITIES

In the Manage Activities Page, the form to create activity is shown in the Fig. No. 6.12, which is used to create the activities and the created activities are displayed in manage activities.



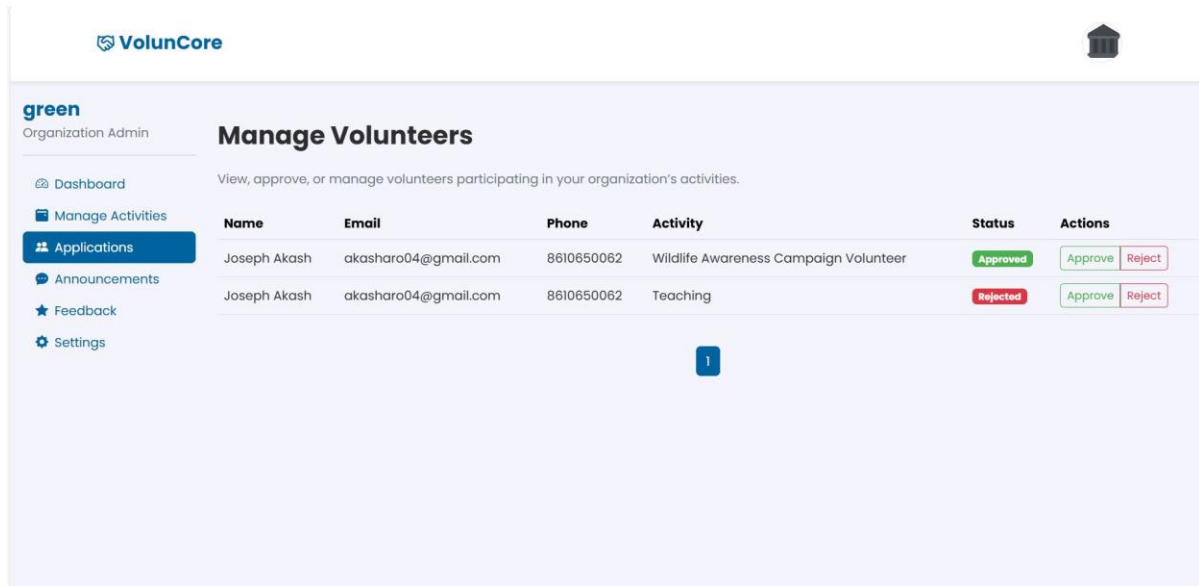
The screenshot displays the 'Post Volunteer Activity' form within the VolunCore interface. The top navigation bar includes the 'VolunCore' logo and a home icon. The left sidebar, under the 'green' organization header, lists navigation options: Dashboard, Manage Activities (highlighted), Applications, Announcements, Feedback, and Settings. The main content area features a 'Back to Manage Activities' link and the form title 'Post Volunteer Activity'. The form fields are organized as follows: 'Organization Name' (text input with 'green' entered) and 'Activity Title' (text input with 'Activity title' entered); 'Description' (text area with 'Activity description' entered); 'Location' (text input with 'Location' entered) and 'Schedule (Date)' (date picker showing 'dd-mm-yyyy'); 'Duration' (text input with 'e.g., 3 hours' entered) and 'Skills Required' (text input with 'e.g., communication, teamwork' entered); 'Applicants Needed' (text input with 'e.g., 5' entered) and 'Mode' (dropdown menu showing 'Select mode'); and a 'Requirements' section at the bottom.

**Fig. No. 6.12 Manage Activities**

In the Manage Activities page, organizations can efficiently create and monitor their volunteer opportunities. The form allows them to input essential details such as the activity title, description, date, time, location, and required number of volunteers. Once submitted, the activity is added to the list below, where all created activities are displayed with key information like status, registered volunteers count, and actions to edit or delete. This page serves as a central space for organizations to manage and track their engagements in a streamlined and user-friendly manner.

## 6.13 APPLICATIONS PAGE

The Applications page is shown in the Fig. No. 6.13, which shows the applications received from the volunteers and make the organization to decide.



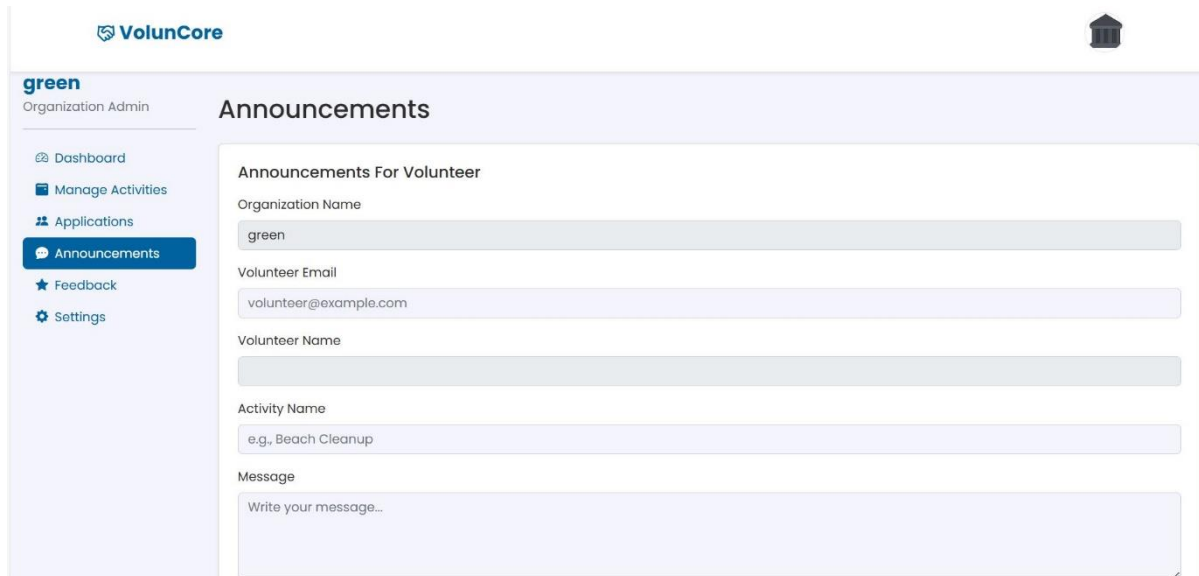
**Fig. No. 6.13 Applications Page**

In the Applications page, the organizations can view and manage all the applications submitted by volunteers for their posted activities. Organizations can easily review these applications and take appropriate action either accepting or rejecting the volunteers based on their suitability for the activity. This page acts as a centralized interface for handling volunteer interest and participation, enabling organizations to make informed decisions efficiently.



## 6.14 ANNOUNCEMENTS PAGE

The Announcements page is shown in the Fig. No. 6.14, which shows the form to make the announcements.



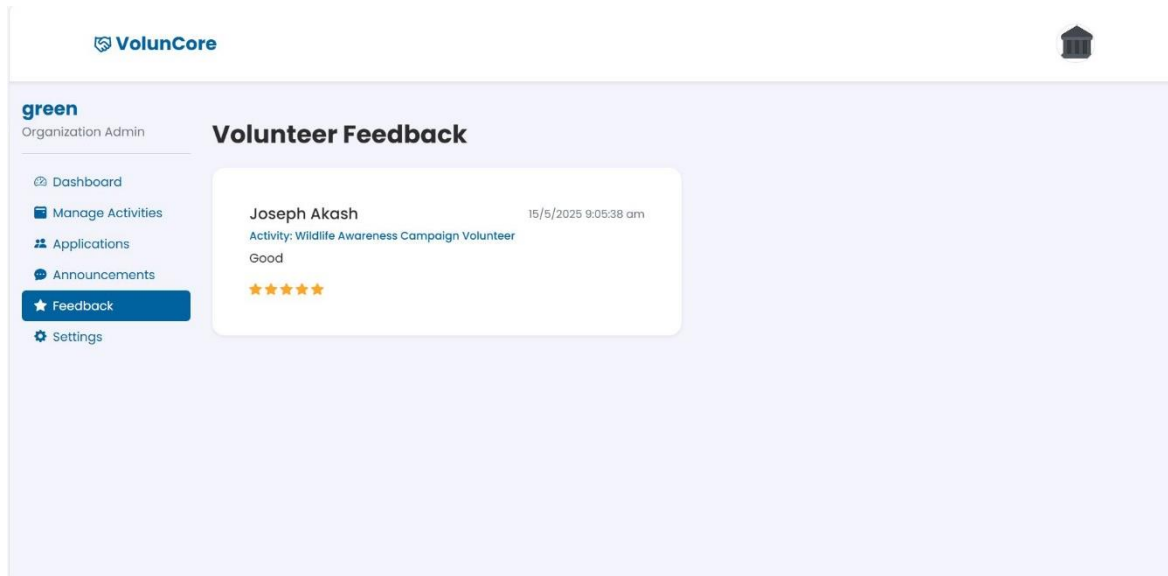
The screenshot displays the 'Announcements' page within the VolunCore system. The interface is for an 'Organization Admin' for an organization named 'green'. On the left, a sidebar menu includes links to Dashboard, Manage Activities, Applications, Announcements (which is highlighted), Feedback, and Settings. The main content area is titled 'Announcements' and contains a form titled 'Announcements For Volunteer'. This form includes input fields for 'Organization Name' (pre-filled with 'green'), 'Volunteer Email' (pre-filled with 'volunteer@example.com'), 'Volunteer Name', 'Activity Name' (with a hint 'e.g., Beach Cleanup'), and a 'Message' field with a placeholder 'Write your message...'. A small house icon is visible in the top right corner of the page header.

**Fig. No. 6.14 Announcements Page**

The Announcements page functions as a simple message board where organizations can post text-based messages intended for volunteers. This page displays only the content of each announcement message along with the date it was posted. It allows organizations to communicate essential updates or information efficiently, and ensures volunteers have a clear, readable list of all relevant announcements in one place. Volunteers can easily stay informed about upcoming events, changes, or important reminders without having to search through multiple channels. This centralized approach helps improve communication and keeps everyone connected.

## 6.15 FEEDBACK VIEW PAGE

The Feedback View page is shown in the Fig. No. 6.15, which shows the feedback collected from the volunteers.



**Fig. No. 6.15 Feedback View Page**

The Feedback View page provides organizations with a consolidated view of all feedback they have submitted for volunteers. Each feedback entry is displayed in a clean, readable format showing details such as the volunteer's name or email, the feedback message, and the date it was submitted and the stars which represented how good the activity was conducted. This page serves as a record-keeping and review interface, allowing organizations to track their past evaluations and maintain transparency. It ensures that all feedback is easily accessible for future reference or audits.

## **CHAPTER – 7**

### **CONCLUSION AND FUTURE WORKS**

#### **7.1 CONCLUSION**

VolunCore is a user-centric platform designed to streamline collaboration between volunteers and nonprofit organizations. It enables organizations to create and manage activities, while volunteers can browse, apply for, and participate in these initiatives based on their interests and availability. The platform offers distinct role-based dashboards, allowing each user to focus on their relevant tasks efficiently. Volunteers can track their participation, submit feedback, and stay informed through announcements, while organizations can manage applications, post updates, and review volunteer performance. The system ensures smooth navigation, secure authentication, and simplified password management to enhance user confidence and trust. With its clean design, real-time updates, and purpose-driven workflows, VolunCore fosters meaningful engagement and encourages long-term community participation.

#### **7.2 FUTURE WORK**

Future enhancements for VolunCore include the development of dedicated mobile applications for Android and iOS, enabling users to stay connected and manage tasks on the go. The introduction of real-time push notifications and dynamic activity mapping will improve user responsiveness and task coordination. Additionally, a comprehensive analytics dashboard will be introduced to provide real-time insights into volunteer engagement, activity progress, and organizational impact. Support for multilingual interfaces is also planned to make the platform more inclusive for diverse user groups. These upcoming features aim to further reduce friction, expand accessibility, and strengthen the collaborative impact between volunteers and organizations.

## **CHAPTER - 8**

### **REFERENCES**

- [1.]** Saleh E., & Shastry, C. A new approach for global task scheduling in volunteer computing systems.(2023). 15(5), (pp.2037–2045).
- [2.]** Alsehaimi&Aref. Impact of Volunteer Work on Improving the Quality of Social Life from the Perspective of Social Work Specialists.(2023).10(1), (pp. 39–61).
- [3.]** Al Ridhawi, I., et al. An incentive-based mechanism for volunteer computing using blockchain. (2021). 20(4), (pp.1–17).
- [4.]** Samanta R, & Ghosh, S K. Sustainable Volunteer Engagement: Ensuring Potential Retention and Skill Diversity for Balanced Workforce Composition in Crowdsourcing Paradigm.(2024).(pp. 1–25).
- [5.]** Sethi B., Samanta R.,&Ghosh S K. Empowering volunteer crowdsourcing services: A serverless-assisted, skill and willingness aware task assignment approach for amicable volunteer involvement. (2022). 10(4), (pp 1234–1247).
- [6.]** Nabil K., & Seghir R. A new social volunteer computing environment with task-adapted scheduling policy. (2021). 13(1),(pp.1–17).
- [7.]** Ihm J., & Shumate M. How volunteer commitment differs in online and offline environments. (2022). 36(4), (pp 583–615).
- [8.]** Waheed A. &Jeon G. An infrastructure-assisted job scheduling and task coordination in volunteer computing-based VANET(2023).9(4), (pp.3613–3633).

# CERTIFICATE



Date: 15.05.2025

## INTERNSHIP CERTIFICATE

This is to certify that **Mr. JOSEPH AKASH C**, Reg.No (962821205026), Student of Dept. of Information Technology, **UNIVERSITY COLLEGE OF ENGINEERING, NAGERCOIL** has successfully completed 3 months Internship in Java Fullstack Development at **Alric Infotech Pvt Ltd**, Trivandrum.

During the period, he has successfully completed a project and also he was methodical and hardworking. He shows a very good analytical and logical approach while studying and developing systems, along with remarkable communication skills.

We wish **Mr. JOSEPH AKASH C** all the best for all the future endeavours.

For Alric Infotech Pvt Ltd

Arun Alfred  
Managing Director

Authorized Signatory



CIN: U72900KL2022PTC077617